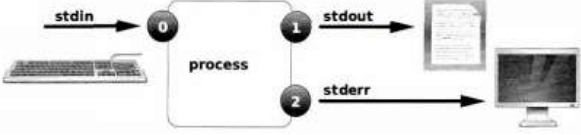
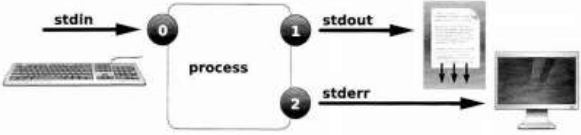
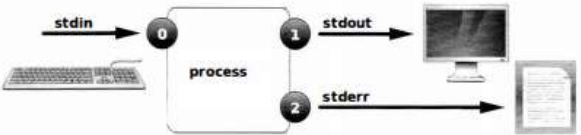
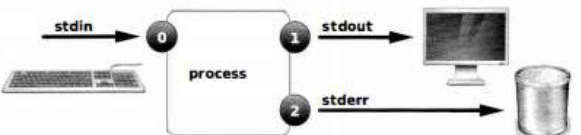
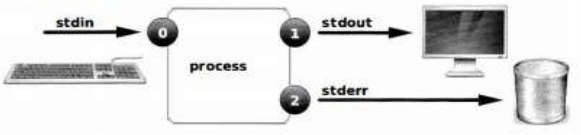
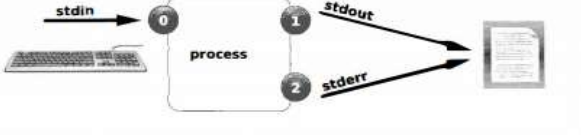
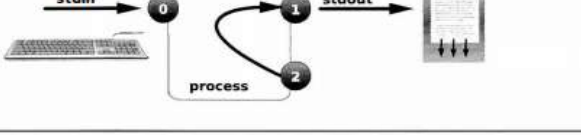


Redirection

Three standard file descriptors :

1. stdin 0 - Standard input to the program.
2. stdout 1 - Standard output from the program.
3. stderr 2 - Standard error output from the program.

<code>>file</code>	redirect stdout to a file ⁽¹⁾	
<code>>>file</code>	redirect stdout to a file, append to current file content ⁽²⁾	
<code>2>file</code>	redirect stderr to a file ⁽¹⁾	
<code>2>/dev/null</code>	discard stderr error messages by redirecting to /dev/null	
<code>2>/dev/null</code>	discard stderr error messages by redirecting to /dev/null	
<code>&>file</code>	combine stdout and stderr to one file ⁽¹⁾	
<code>>>file 2>&1</code>	combine stdout and stderr , append to current file content ^{(2) (3)}	

Examples for output redirection

- Save a timestamp for later reference.

[student@desktopX -]\$ date > /tmp/saved - timestamp

- Copy the last 100 lines from a log file to another file.

```
[student@desktopX -]$ tail -n 100 /var/log/dmesg > /tmp/latest - 100 - boot - messages
```

- Concatenate four files into one.

```
[student@desktopX -]$ cat file1 file2 file3 file4 > /tmp/all-four-in-one
```

- List the home directory's hidden and regular file names into a file.

```
[student@desktopX -]$ ls -a > /tmp/myfile
```

- Append output to an existing file.

```
[student@desktopX -]$ echo "new line of information" >> /tmp/many
```

- In the next examples, errors are generated since normal users are denied access to system directories. Redirect errors to a file while viewing normal command output on the terminal.

```
[student@desktopX -]$ find /etc -name passwd 2> /tmp/errors
```

- Save process output and error messages to separate files.

```
I [student@desktopX -]$ find /etc -name passwd > /tmp/output 2> /tmp/errors
```

- Ignore and discard error messages.

```
[student@desktopX -]$ find /etc -name passwd > /tmp/output 2> /dev/null
```

- Store output and generated errors together.

```
[student@desktopX -]$ find /etc -name passwd &> /tmp/both
```

- Append output and generated errors to an existing file.

```
[student@desktopX -]$ find /etc -name passwd >> /tmp/save - both 2>&1
```

Constructing pipelines

Redirection controls channel output to or from *files* while *piping* sends channel output to another process.

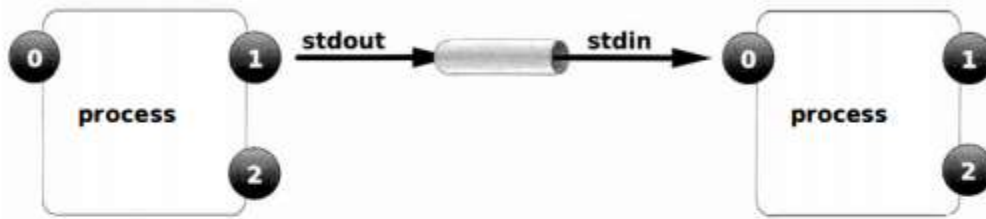


Figure 4.8: Process I/O piping

Examples for process piping redirection

Paginate a command's long output.

```
[student@desktopX ~]$ ls -l /usr/bin | less
```

Count the number of lines in an output or listing.

```
[student@desktopX ~]$ ls | wc -l > /tmp/how-many-files
```

Grab the first lines, last lines, or selected lines of command output.

```
[student@desktopX ~]$ ls -t | head -n 10 > /tmp/ten-last-changed-files
```

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